

CalorieSense

AI-Powered Calorie Prediction & Adaptive Fitness Coaching

Final Project Presentation – Group 11



TEAM MEMBERS

D.M.I.N Dissanayake

KIC-HNDCSAI252F-003

C.A Aneek Ahamed

KIC-HNDCSAI252F-012

M.K Imadh Ahamed

KIC-HNDCSAI252F-032

M.H. Nasrullah

KIC-HNDCSAI252F-031

M.M Abdullah

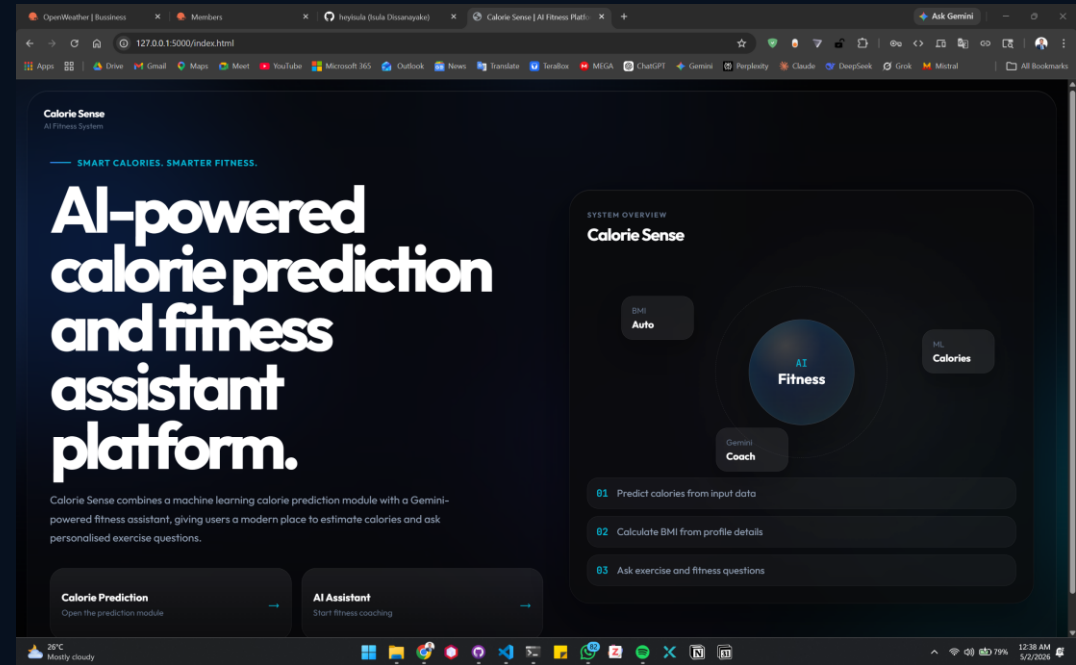
KIC-HNDCSAI252F-039

The gap CalorieSense solves

WHY THIS PROJECT MATTERS

Many fitness tools still estimate calories using generic formulas. CalorieSense was built to make calorie prediction more personalised, context-aware, and useful in real fitness sessions.

- Traditional estimators often ignore age, weight, gender and real exercise intensity.
- Weather and environment are normally not part of calorie estimation.
- Most apps stop at numbers; users still need practical exercise guidance.



Dataset strategy

WHAT EACH DATA SOURCE DOES

1	Exercise & Fitness Metrics	ANN training data
2	ExerciseDB	Exercise library for Coaching and Recommendation
3	Mendeley Gym Recommendation	Exercise library for Coaching and Recommendation
4	OpenWeatherMap API	Live weather context

Model Training Dataset: Dataset 2 only, because the planned merge created incompatible calorie distributions and many missing values.

3,864 training rows **4 data sources** **80/20 train-test split**

A hidden data problem was discovered

THE ORIGINAL TARGET COULD NOT BE LEARNED

Early models failed because the original Calories Burn column behaved like random synthetic noise.

- Feature correlations were almost zero before fixing the target.
- KS test showed the target behaved like a uniform random distribution.
- This explained the early negative or near-zero R^2 results.

Before fix (R^2)

Duration +0.02
Heart Rate -0.04
Intensity +0.01

After fix (R^2)

Duration +0.67
Heart Rate +0.59
Intensity +0.18

The target was rebuilt scientifically

KEYTEL FORMULA + REALISTIC ADJUSTMENTS

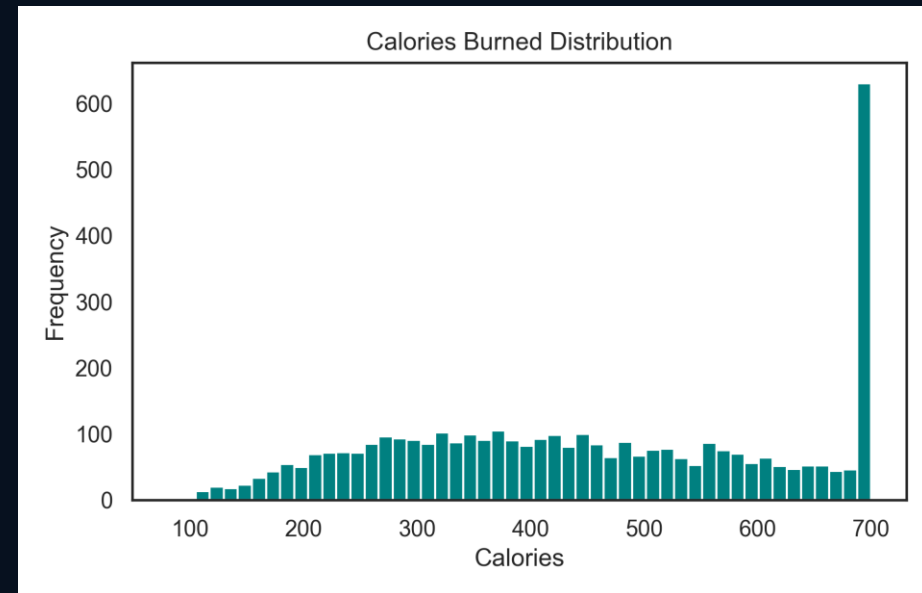
Instead of accepting a random target, the project rebuilt calories using a validated heart-rate based energy expenditure equation.

Male / Female formula uses:

- Duration
- Heart Rate
- Weight
- Age
- Gender

+ intensity adjustment +
realistic Gaussian noise

```
male_cal = durMin * (-55.0969 + 0.6309*hr + 0.1988*weight + 0.2017*age) / 4.184  
female_cal = durMin * (-20.4022 + 0.4472*hr - 0.05741*weight + 0.074*age) / 4.184
```



Critical fix: Calories column was rebuilt

THE ORIGINAL TARGET COULD NOT BE LEARNED

01 Problem found

The original Calories Burn column behaved like random synthetic noise. Early models produced near-zero / negative R^2 because there was no meaningful relationship between inputs and target.

02 Scientific replacement

We removed the unreliable target and rebuilt calories using the validated Keytel et al. heart-rate formula using duration, heart rate, weight, age and gender.

03 Final result

The rebuilt target created real physiological signal. Duration, heart rate and intensity became meaningful predictors, allowing the model to learn properly.

Male/Female Keytel formula → intensity adjustment → weather multiplier → final Calories Burned target

Weather-aware calorie prediction

ADDING REAL-WORLD CONTEXT

Weather is fetched live and used as a conservative physiological adjustment at inference time.

Sunny

+5%

heat dissipation strain

Cloudy

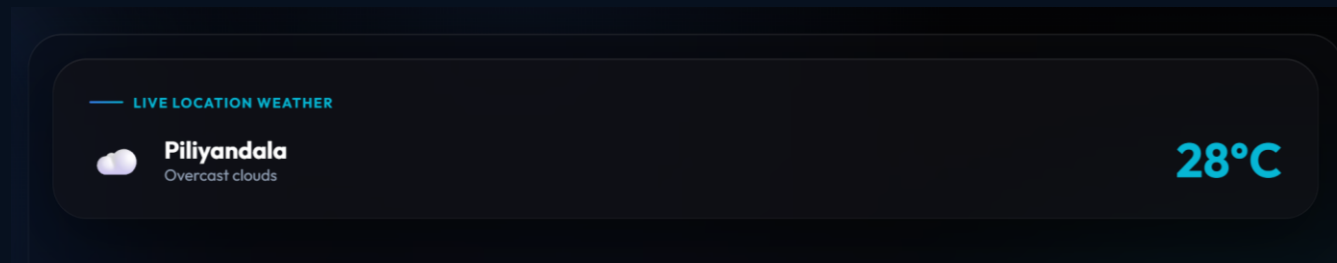
0%

baseline condition

Rainy

+8%

cold-induced thermogenesis



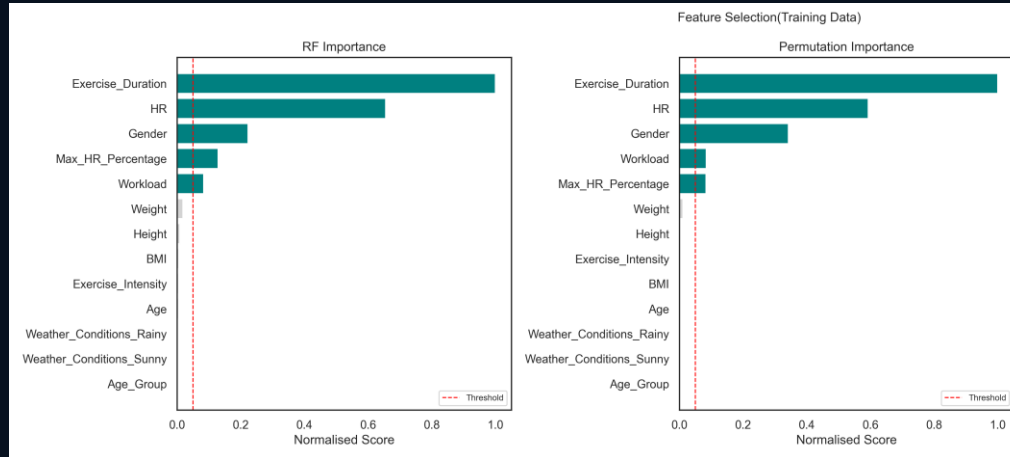
Leakage-free preprocessing pipeline

FROM RAW DATA TO MODEL INPUTS

-
- | | | | | | |
|----|------------------|----|---------------------------------|----|-----------------------|
| 01 | Load raw data | 02 | Drop non-predictive columns | 03 | Convert duration |
| 04 | Rebuild target | 05 | Encode gender + weather | 06 | Engineer features |
| 07 | Train/test split | 08 | Feature selection on train only | 09 | Scale train then test |

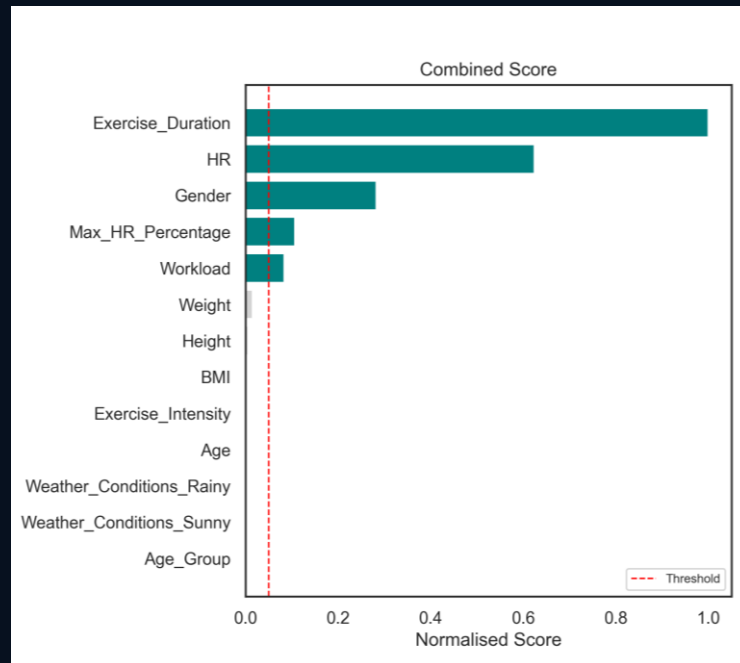
Feature selection confirmed the strongest signals

DURATION AND HEART RATE DOMINATE



Selected 7 features from 13 candidates using Random Forest importance and Permutation Importance on training data only.

- Exercise Duration was the strongest feature.
- Heart Rate was the second strongest signal.
- Weather has low importance by design, but still affects live inference.



Models trained and evaluated

SIX FAIR COMPARISONS ON THE SAME FINAL PIPELINE

Baseline

Ridge Regression

Polynomial interaction baseline for simple explainable comparison.

Untuned

Random Forest

Tree ensemble model trained before tuning.

Tuned

Random Forest tuned

RandomizedSearchCV tuning with 5-fold validation.

Untuned

XGBoost

Gradient boosting model trained before tuning.

Tuned

XGBoost tuned

Strongest competitor, but slightly larger generalisation gap.

Selected

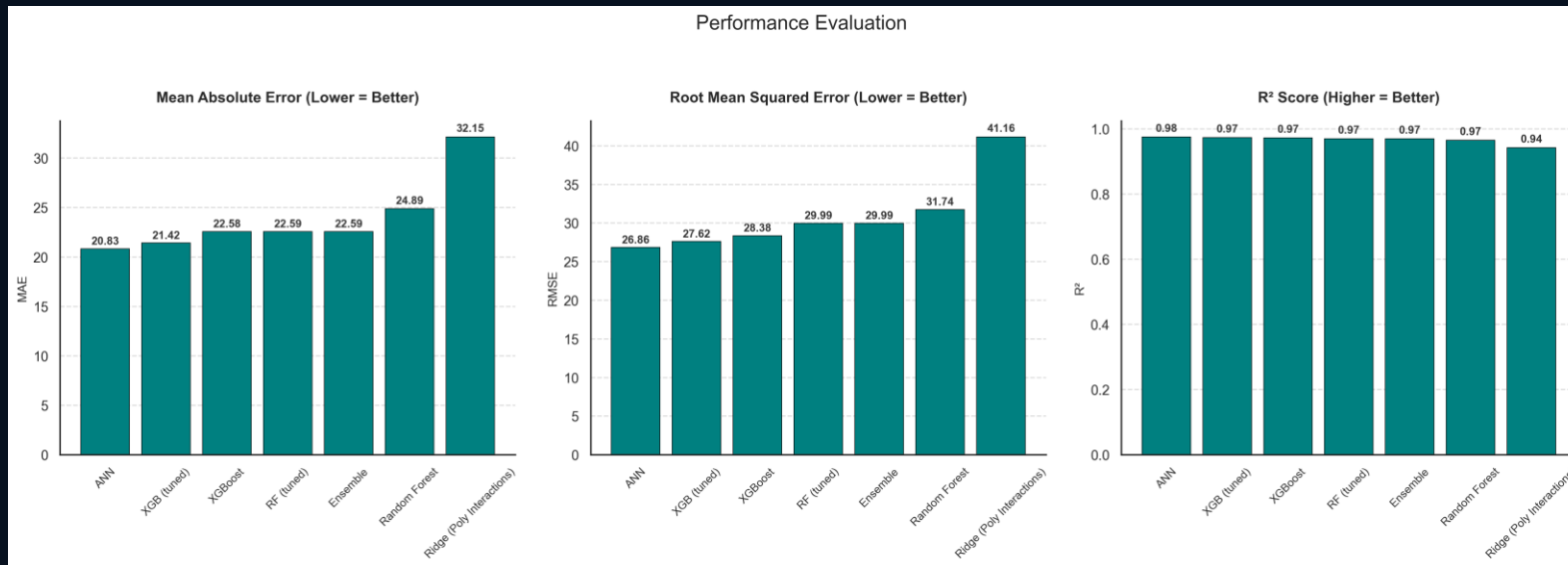
ANN

Artificial Neural Network with dense layers, BatchNorm, Dropout and Huber loss.

All models were tested after the target rebuild, feature engineering, feature selection and scaling steps.

Model comparison results

ANN SELECTED FOR BEST GENERALISATION



0.9757
ANN Test R^2

20.83
ANN Test MAE

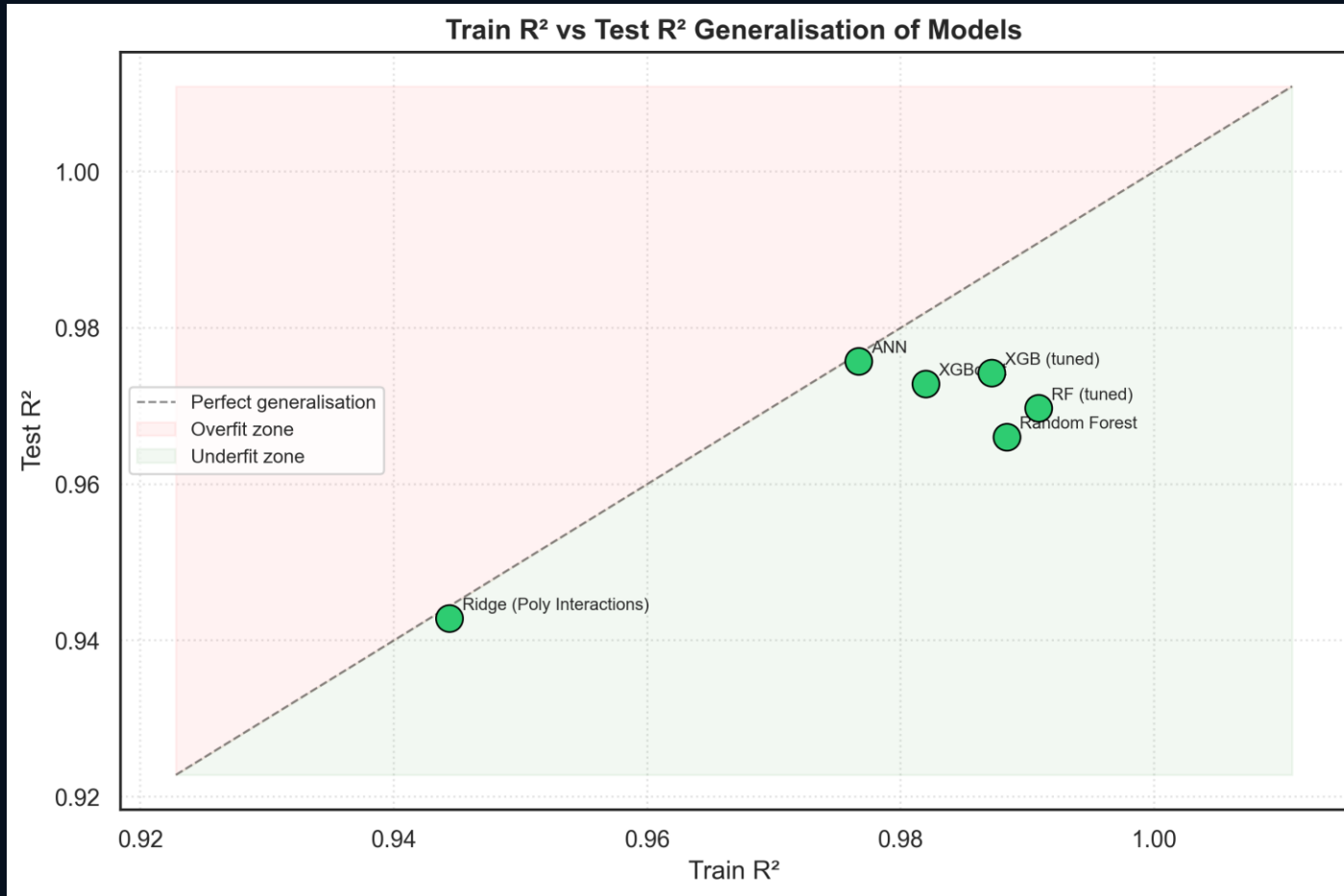
0.0011
Train/Test gap

26.86
Test RMSE

The ANN produced the most balanced result: high accuracy with the smallest generalisation gap across all tested models.

Why ANN over tuned XGBoost?

GENERALISATION MATTERED MORE THAN TRAINING SCORE



XGBoost tuned had slightly stronger training R², but the ANN had a much smaller train/test gap. For live predictions, stable performance on unseen users is more important.

0.0130
XGB tuned ΔR^2

0.0011
ANN ΔR^2

Selected model: Artificial Neural Network

CHOSEN FOR STRONGEST GENERALISATION IN LIVE PREDICTION

0.9757

Test R^2

20.83 kcal

Test MAE

0.0011

ΔR^2 gap

A Why not only choose the highest train score?

Tuned XGBoost had a slightly higher training R^2 , but its train/test gap was much larger. For real users, every input is unseen data, so stable generalisation is more important than memorising training patterns.

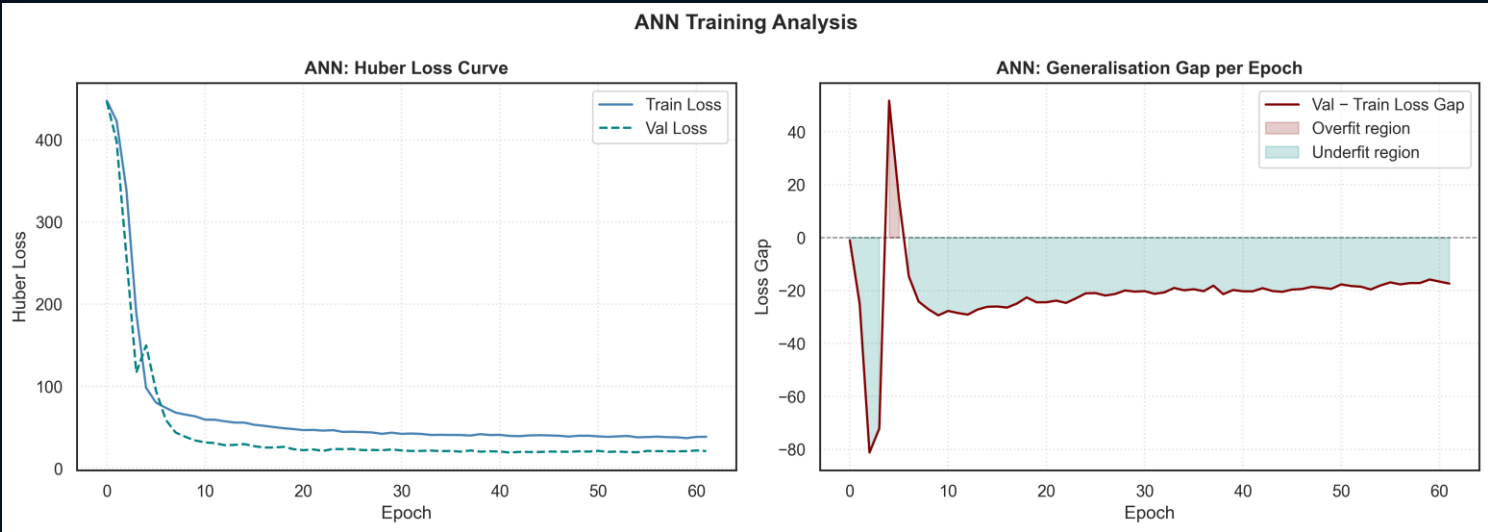
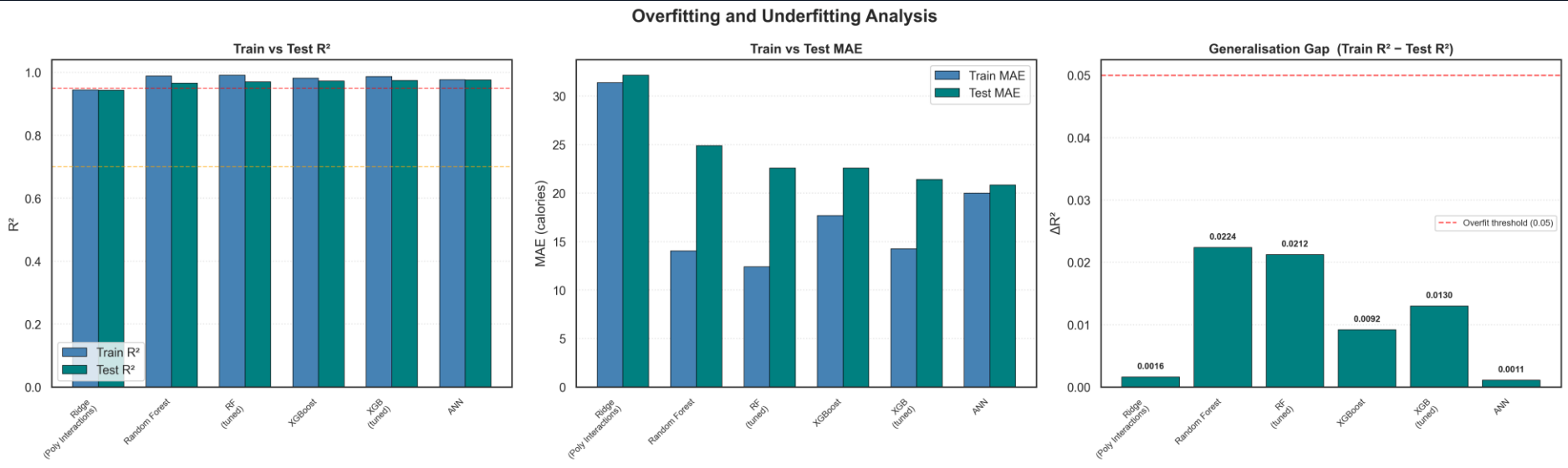
B Why ANN was the final production model

The ANN delivered almost the same high accuracy while keeping the smallest train/test gap. This made it the safest option for deployment inside the Flask prediction API.

Final decision: ANN gave high accuracy with the smallest generalisation gap.

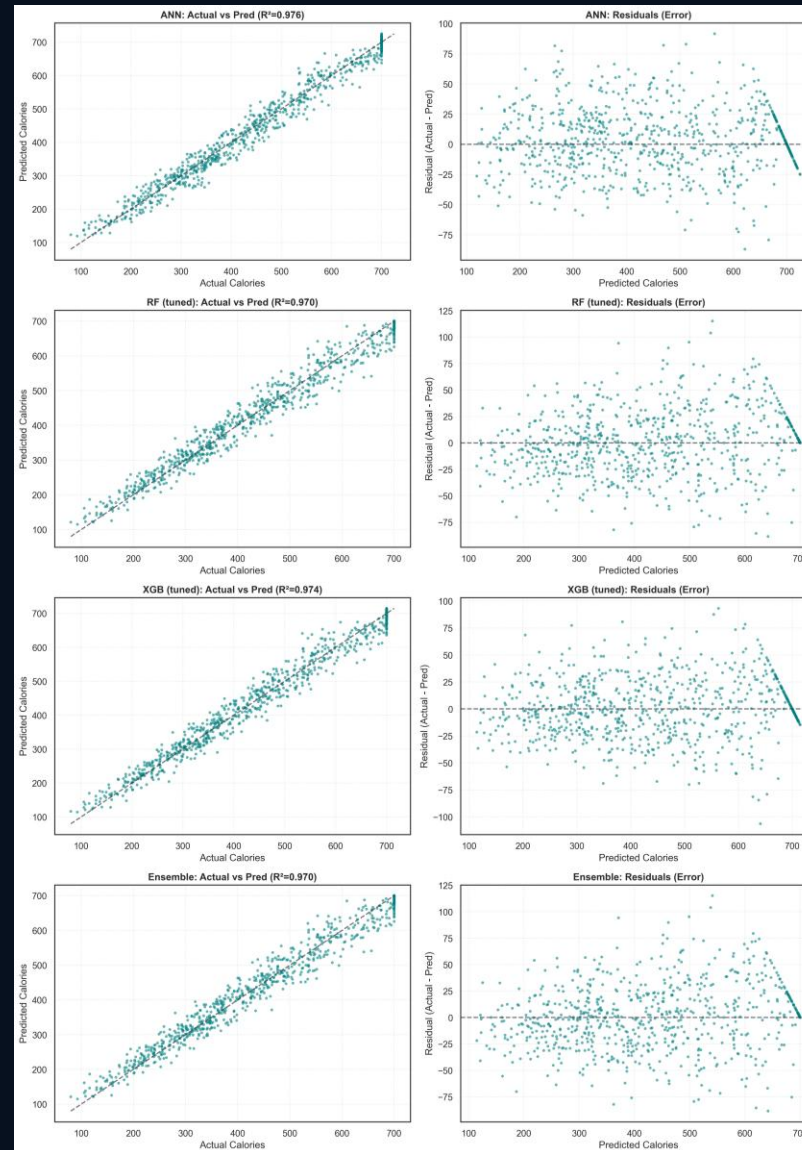
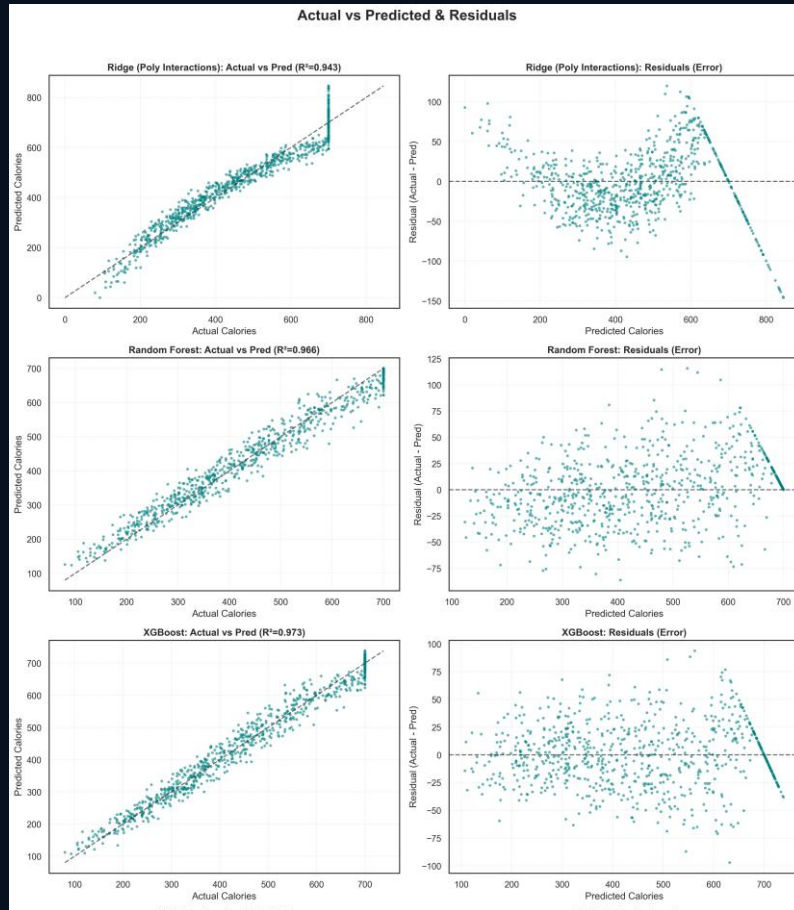
ANN training diagnostics and Overfitting Underfitting Analysis

CLEAN CONVERGENCE WITHOUT OVERFITTING



Prediction quality

ACTUAL VS PREDICTED CALORIES

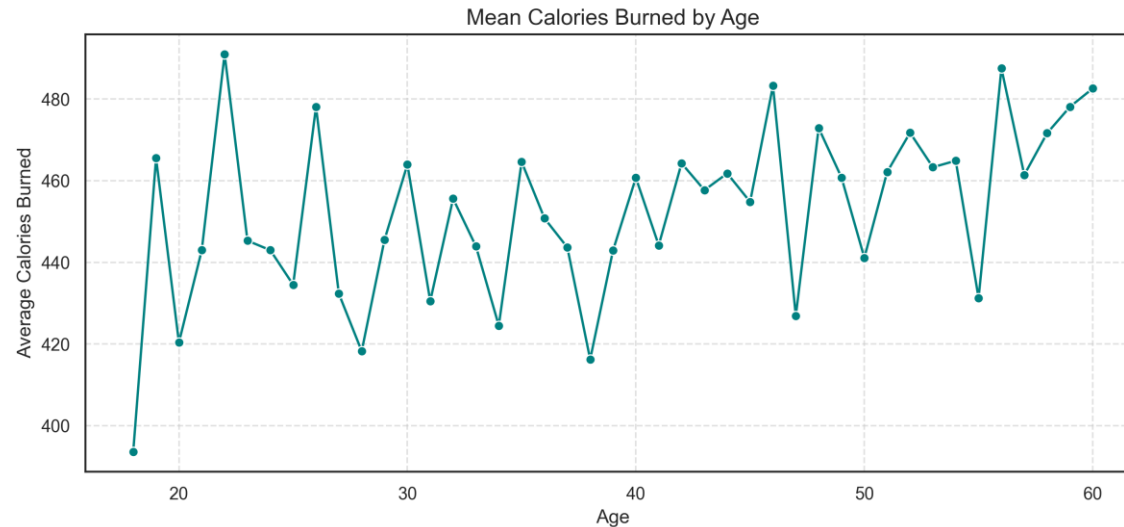


Predictions stay close to the ideal line, and residuals are centred near zero. This supports the claim that the ANN is not systematically over- or under-estimating.

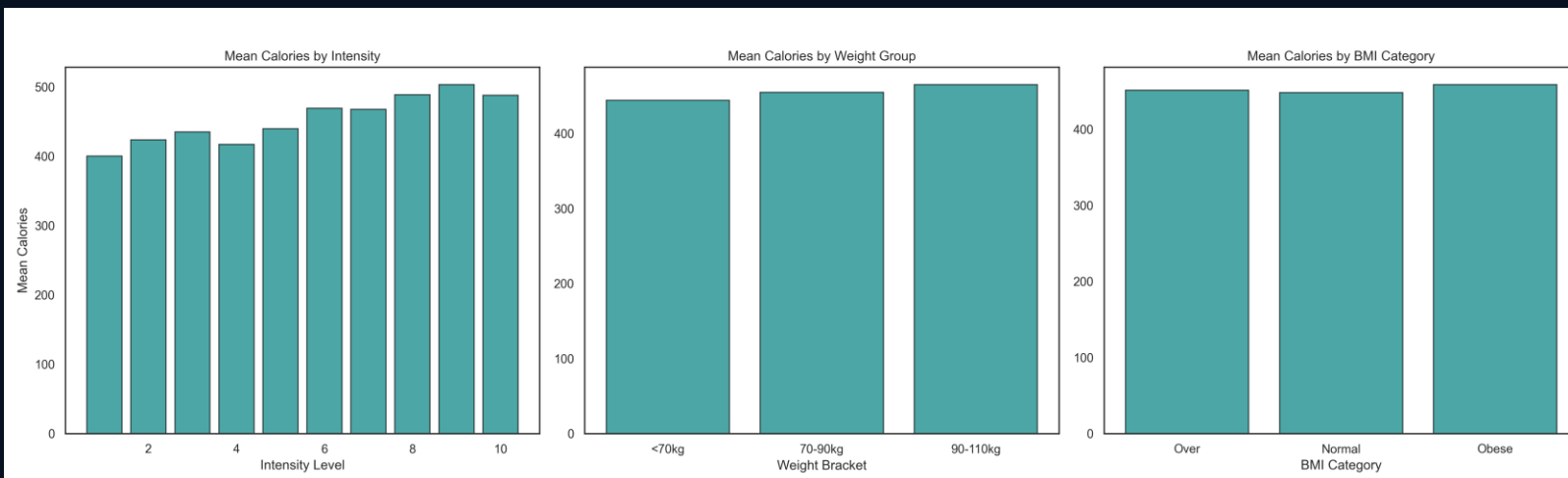
- Good alignment between actual and predicted values.
- Residuals are approximately balanced around zero.
- Model behaviour is explainable and deployment-ready.

Exploratory data analysis highlights

DATA BEHAVIOUR AFTER THE FIX

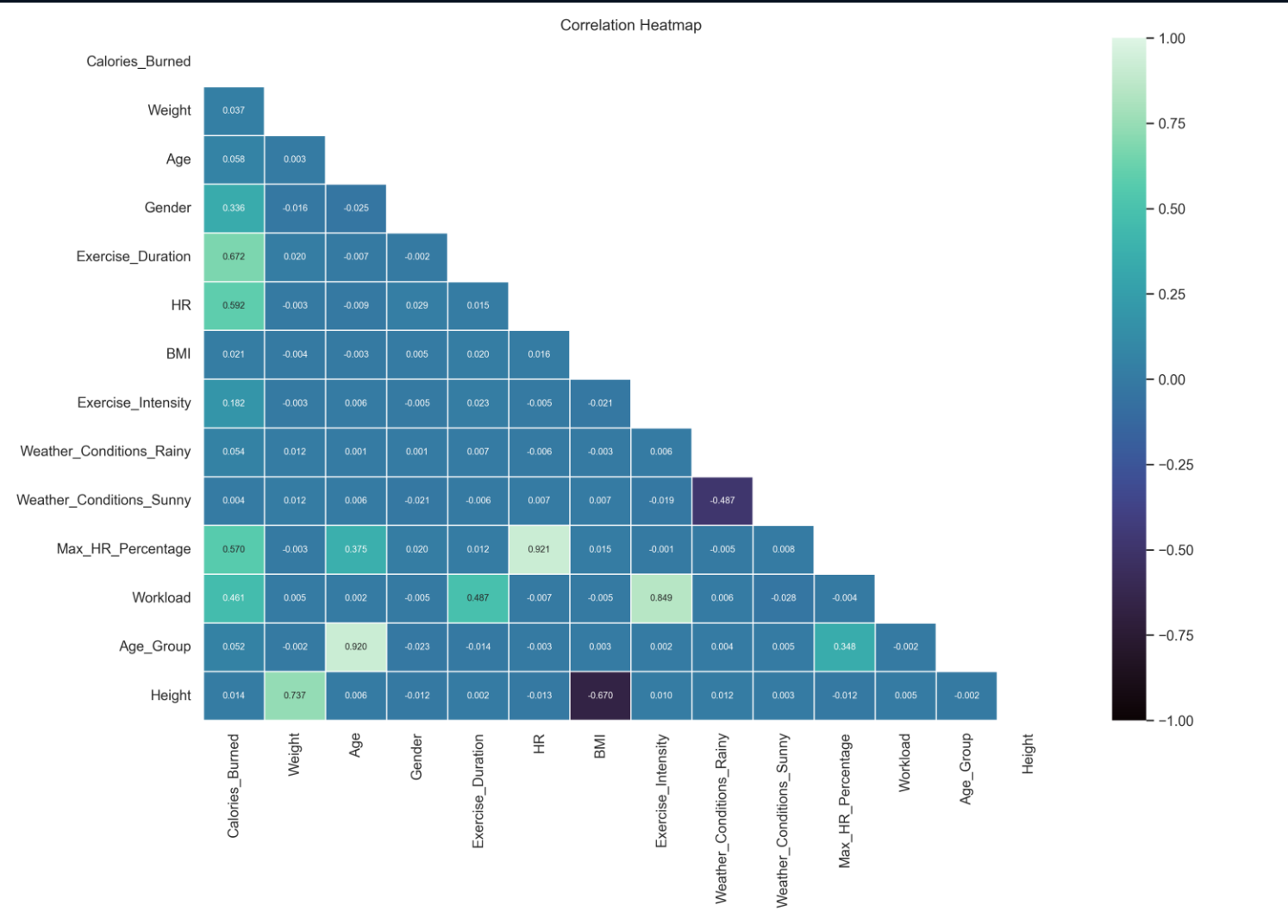


EDA confirmed the target became meaningful after the Keytel replacement. Duration and heart rate became the strongest practical indicators of calorie burn.



Exploratory data analysis highlights

DATA BEHAVIOUR AFTER THE FIX



EDA confirmed the target became meaningful after the Keytel replacement. Duration and heart rate became the strongest practical indicators of calorie burn.

Calorie Prediction Workflow

FROM FORM INPUT TO KCAL OUTPUT

Calorie Sense
Prediction Module

ENTER EXERCISE DATA

GENDER
Male

AGE
25

HEART RATE (BPM)
120

EXERCISE DURATION (HOURS)
1.0

EXERCISE INTENSITY (1-10)
5

Predict Calories Burned

LIVE LOCATION WEATHER
Piliyandala
Overcast clouds
28°C

Awaiting Input
Fill out the form and predict to see calories burned.

1. User enters workout details

2. Weather is fetched automatically

3. Derived features are computed

Calorie Sense
Prediction Module

ENTER EXERCISE DATA

GENDER
Male

AGE
22

HEART RATE (BPM)
119

EXERCISE DURATION (HOURS)
0.2

EXERCISE INTENSITY (1-10)
2

Predict Calories Burned

LIVE LOCATION WEATHER
Piliyandala
Overcast clouds
28°C

PREDICTED ENERGY EXPENDITURE
116.70
kcal burned

4. Scaler prepares the feature vector

5. ANN predicts calorie expenditure

6. Result appears in the same page

Key challenges faced

WHAT MADE THE PROJECT REALISTIC

Random target

The original Calories Burn column had no learning signal.

Dataset merge issue

Two calorie distributions and many null values made merging unreliable.

Leakage prevention

Pipeline order had to be corrected across iterations.

Weather importance

Low feature score needed careful scientific explanation.

API reliability

Weather and Gemini integration required fallback handling.

Conclusion

FINAL MESSAGE

CalorieSense is more than a prediction demo. It shows a full AI development journey: finding a broken target, fixing it with scientific reasoning, comparing models fairly, and deploying a working fitness platform.

**ANN
production
model**

**95.38%
Prediction
Accuracy**

**Gemini
AI coach**

**Weather
context-aware**

Thank you